

Detected by:-
Semantic Analyzer.

Handling:-

- Report meaningful error messages.
- Stop compilation (usually).

Eg:- `int a;`
`a = "hello";` // type mismatch

`int b;`
`print(c);` // undeclared variable

9) Loop optimization improves performance by reducing execution time inside loops, since loop runs many times.

a) Loop Invariant Code Motion:-
Move computations outside the loop if they do not change during iterations.

Benefit:-

- Avoid repeated computation.
- Improves efficiency.

Eg:- Before:- `for (i = 0; i < n; i++) {`
`x = a + b;`
`y = x * i;`
`}`

After optimization:- `x = a + b;`
`for (i = 0; i < n; i++) {`
`y = x * i;`
`}`